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SBL Trigger Logic and Proposed Revisions

1. Current Baseline Methodology

Calculate a threshold (T_p) for each product using the historical "1A K-lot" (fail rate) distribution.

Formula:

$$T_p = \mu + (Z \times \sigma)$$

Where:

- μ **(Mu):** The Average 1A K-lot fail rate for this product (last 90-180 days).
- σ **(Sigma):** The Standard Deviation of the 1A K-lot fail rate.

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{N}}$$

- Z **(Sensitivity Factor):** A variable multiplier determined by how recoverable the specific product is.

Determining the 'Z' Score (The Efficiency Lever)

Adjust the strictness of the trigger based on the product's **Historical Recovery Rate** (from "Details of LOH" data).

- **High Recovery Rate (>80%):** Mostly false failures or retest-recoverable -> need these to flow naturally to standard retest (Test 2A) rather than holding the lot $\rightarrow$ **Higher Threshold (Looser)**.
- **Low Recovery Rate (<50%):** High failures likely indicate true defects $\rightarrow$ **Lower Threshold (Stricter)** to hold lots early for inspection or scrapping.

Z-Score Assignment Matrix:

Historical Recovery Rate	Category	Recommended Z-Score	Effect on Threshold
> 80%	**Safe / High False Fail**	**4-5**: depends on margin needed	**High / Loose**: maximizes flow to Test 2A.
50% - 80%	**Standard**	**3**	**Normal**: filters meaningful statistical outliers.
< 50%	**Risky / Real Defects**	**2**	**Low / Strict**: even small spikes trigger a Hold.

Example Implementation

Details of LOH

Total trigger of LOH last 360 days

Row Labels	#lot	volume	Average of Klot	Sum of time_on_hold	Sum of testing_time
MVOU	**91**	**134063**	**13.73**	**29.9**	**23.5**
RPR8161	50	77798	11.87	11.9	4.7
RPL8161	24	36374	18.93	11.7	17.3
KBZ	5	8071	13.38	0.7	0.8
SKA	3	3212	10.95	0.2	0.1
CMLS62	2	2938		0.2	0.2
CFD	2	3444	11.85	0.1	0.2
RPLPS6	1	96	13.79	0.6	0.0
CFA	1	224		0.6	0.0
CMLS102	1	210	27.62	3.7	0.0
GRR	1	130		0.2	0.1
KBLU22	1	1566		0.1	0.0
TRST	**113**	**167896**	**2.99**	**32.7**	**22.3**
RPR8161	46	71944	1.26	12.4	5.2
RPL8161	15	23436	1.34	5.5	9.1
WHLU42	11	17316	4.70	1.9	0.9
CMLS102	10	16916	9.21	1.7	1.7
KBLU22	9	11041		0.9	1.1
RPLP682	9	8698	0.00	2.3	0.7
RRC8161	4	6272		3.0	2.6
CFD	3	5161	2.27	0.9	0.3
CFA	2	1946		0.1	0.1
RPG8161	1	1568		0.6	0.3
KBZ	1	1722	3.31	0.3	0.1
RPLPS6	1	84	0.00	0.2	0.0
RPRP682	1	1792		2.9	0.2
Grand Total	**204**	**301959**	**7.38**	**62.6**	**45.8**

Recovery rate base on last 180 days historical data

	Release		Retest		
Row Labels	Count of lot	Average of Klot	Count of lot	Average of Klot	Recovery_rate
RPR8161	50	11.87	46	1.26	**83%**
RPL8161	24	18.93	15	1.34	**78%**
WHLU42			11	4.70	**87%**
CMLS102	1	27.62	10	9.21	**67%**
KBLU22	1		9		**83%**
RPLP682			9	0.00	**92%**
KBZ	5	13.38	1	3.31	**87%**
CFD	2	11.85	3	2.27	**60%**
RRC8161			4		**No Data**
SKA	3	10.95			**90%**
CFA	1		2		**45%**
RPLPS6	1	13.79	1	0.00	**No Data**
CMLS62	2				**89%**
GRR	1				**84%**
RPRP682			1		**91%**
RPG8161			1		**93%**

Most of products have recovery rate higher than 80% at 2A

Product RPL8161

- **Current State:**
  - Average K-lot: **18.93%**
  - Recovery Rate: **78%** (Standard)
  - Current Trigger limit: **15%** (constant)

Recovery rate base on last 180 days historical data

Row Labels	Release Count of lot	Release Average of Klot	Retest Count of lot	Retest Average of Klot	Recovery_rate
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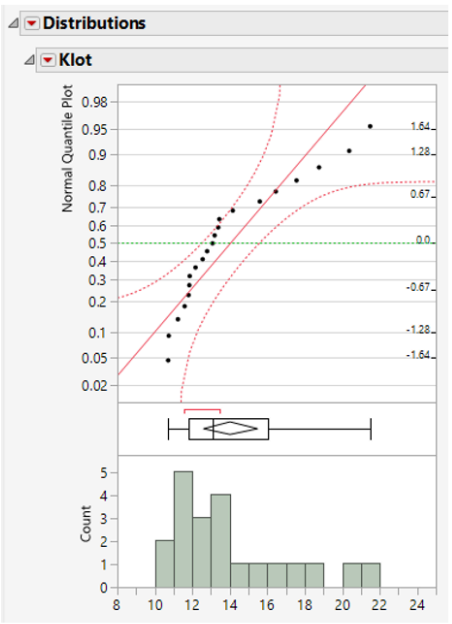
KBLU22	1		9		**83%**
RPLP682			9	0.00	**92%**
KBZ	5	13.38	1	3.31	**87%**
CFD	2	11.85	3	2.27	**60%**
RRC8161			4		**No Data**
SKA	3	10.95			**90%**
CFA	1		2		**45%**
RPLPS6	1	13.79	1	0.00	**No Data**
CMLS62	2				**89%**
GRR	1				**84%**
RPRP682			1		**91%**
RPG8161			1		**93%**

- **Proposed Baseline State (Estimated):**
  - The Standard Deviation (σ) is roughly **3.07%**.

RPL8161 trigger on hold high klot

Disposition: release (no retest)



lot	1A Klot	recovery_rate_predicted	final_klot_predicted
U408P820	12.79	10.45	2.34
U440D822	21.49	18.23	3.26
U440D844	20.36	17.78	2.58
U441D936	16.44	13.82	2.62
U442F541	18.76	15.99	2.77
U444B549	11.58	8.74	2.84
U444D146	11.87	9.66	2.21
U446D884	11.79	9.36	2.43
U446F545	14.15	11.36	2.79
U446G255	10.71	7.91	2.80
U449H304	11.20	8.79	2.41
U450H154	12.18	9.55	2.63
U451C063	13.38	11.04	2.34
U451G625	17.58	15.69	1.89
U503B649	10.74	7.59	3.15
U503B672	13.01	9.95	3.06
U503C020	15.61	11.88	3.73
U504C113	13.44	9.69	3.75
U504C514	11.83	8.58	3.25
U504D419	13.18	9.02	4.16
U507C047	12.57	8.64	3.93
		233.72	

- All of lots is predicted to have high recovery rate
- If all lots retest 2A, the predicted k-lot units could be saved is 233% x 1.4 ku = **3.2 ku**

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- Since Recovery is **78%** $\rightarrow Z = 3$
- **New Threshold:** $18.93\% + (3 \times 3.07\%) = 28.14\%$

- **Re-implementing the new threshold on previous fail lots:**

lot	1A Klot
U408P820	12.79
U440D822	21.49
U440D844	20.36
U441D936	16.44
U442F541	18.76
U444B549	11.58
U444D146	11.87
U446D884	11.79
U446F545	14.15
U446G255	10.71
U449H304	11.20
U450H154	12.18
U451C063	13.38
U451G625	17.58
U503B649	10.74
U503B672	13.01
U503C020	15.61
U504C113	13.44
U504C514	11.83
U504D419	13.18
U507C047	12.57

Lot ID	1A Klot (Fail %)	Current Status (15% Limit)	Proposed Status (28.14% Limit)	Result
U440D822	**21.49%**	**HOLD**	**RELEASE (to Test 2A)**	**Saved**
U440D844	**20.36%**	**HOLD**	**RELEASE (to Test 2A)**	**Saved**
U442F541	**18.76%**	**HOLD**	**RELEASE (to Test 2A)**	**Saved**
U441D936	**16.44%**	**HOLD**	**RELEASE (to Test 2A)**	**Saved**
U451G625	**17.58%**	**HOLD**	**RELEASE (to Test 2A)**	**Saved**
U503C202	**15.61%**	**HOLD**	**RELEASE (to Test 2A)**	**Saved**

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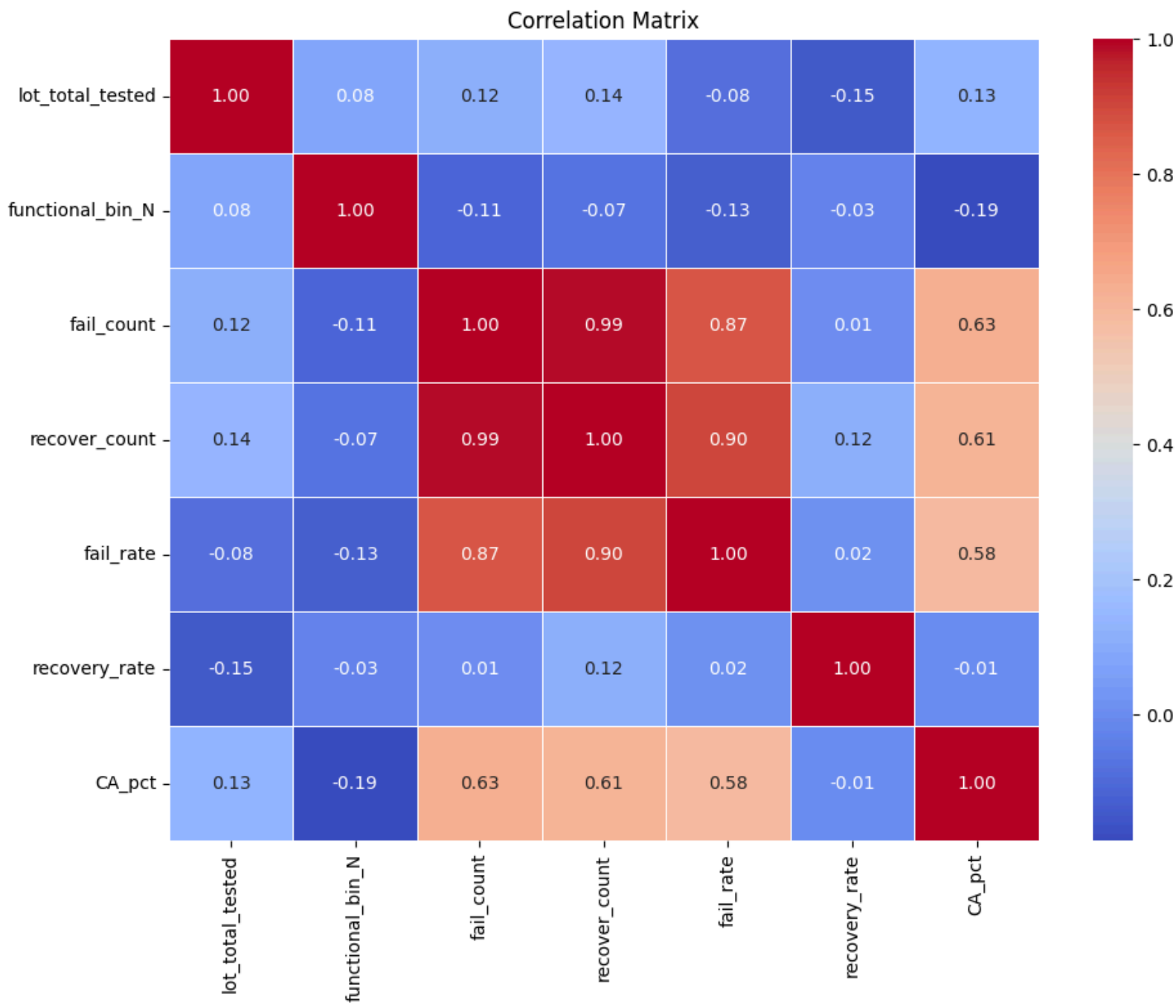
2. Problems

Data Findings and Issues:

1. **Hardware failures skip the queue (Hardware failure not detected):**
   - SBL looks at `klot` % (fail rate). It ignores `CA_pct` (concentrated failure).
   - *Insight from 90-day dataset:* Rows with **CA_pct $\leq$ 0.05** (concentrated hardware failures) make up **54% of all records** and possess a **93.1% average structural recovery rate** when fixed, compared to 83.3% for other failures.


Correlation Matrix
lot_total_tested
functional_bin_N
fail_count
recover_count
fail_rate
recovery_rate
CA_pct
1.00 0.08 0.12 0.14 -0.08 -0.15 0.13
0.08 1.00 -0.11 -0.07 -0.13 -0.03 -0.19
0.12 -0.11 1.00 0.99 0.87 0.01 0.63
0.14 -0.07 0.99 1.00 0.90 0.12 0.61
-0.08 -0.13 0.87 0.90 1.00 0.02 0.58
-0.15 -0.03 0.01 0.12 0.02 1.00 -0.01
0.13 -0.19 0.63 0.61 0.58 -0.01 1.00
1.0
0.8
0.6
0.4
0.2
0.0

Downloads > PRJS > LCBI_SmartDisposition > files > final_all_product_90days(in).csv

```csv
,DLCP,lot,lot_total_tested,functional_bin_N,fail_count,recover_count,fail_rate,recovery_rate,CA_pct
P,U450G425,1568,804,2,0,0.0013,0,0.0312
P,U450G425,1568,827,2,2,0.0013,1,0.0625
P,U450G425,1568,837,34,28,0.0217,0.82,0.25
P,U450G425,1568,838,1,0,0.0006,0,0.0312
P,U450G425,1568,1930,2,2,0.0013,1,0.0625
P,U450G425,1568,2721,10,10,0.0064,1,0.0312
P,U450G425,1568,9984,1,1,0.0006,1,0.0312
P,U450G425,1568,9987,1,1,0.0006,1,0.0312
P,U450G425,1568,9991,1,1,0.0006,1,0.0312
P,U450G474,1318,804,6,4,0.0046,0.67,0.0625
P,U450G474,1318,827,1,1,0.0008,1,0.0312
P,U450G474,1318,837,53,52,0.0402,0.98,0.2188
P,U450G474,1318,838,1,1,0.0008,1,0.0312
P,U450G474,1318,1939,1,1,0.0008,1,0.0312
P,U450G474,1318,9764,1,1,0.0008,1,0.0312
P,U450G474,1318,9766,2,2,0.0015,1,0.0625
P,U450G545,1568,827,7,7,0.0045,1,0.125
P,U450G545,1568,837,1,1,0.0006,1,0.0312
P,U450G545,1568,838,1,0,0.0006,0,0.0312
P,U450G545,1568,1930,2,2,0.0013,1,0.0312
P,U450G545,1568,1939,1,1,0.0006,1,0.0312
P,U450G675,1316,827,4,3,0.003,0.75,0.0625
P,U450G675,1316,837,5,4,0.0038,0.8,0.0938
P,U450G675,1316,1930,1,1,0.0008,1,0.0312
P,U450G675,1316,1939,2,1,0.0015,0.5,0.0625
P,U450G675,1316,9984,7,1,0.0053,0.14,0.0312
P,U450G629,1568,827,3,3,0.0019,1,0.0625
P,U450G629,1568,828,1,0,0.0006,0,0.0312
P,U450G629,1568,837,9,9,0.0057,1,0.0938
P,U450G629,1568,1939,1,1,0.0006,1,0.0312
P,U450G629,1568,9764,1,1,0.0006,1,0.0312
P,U450G629,1568,9765,1,1,0.0006,1,0.0312
P,U450G730,1317,827,1,1,0.0008,1,0.0312
P,U450G730,1317,837,6,6,0.0046,1,0.0625
P,U450G730,1317,1930,1,1,0.0008,1,0.0312
P,U450G730,1317,1939,2,2,0.0015,1,0.0938
V,U450G287,1050,827,1,1,0.001,1,0.0312
V,U450G287,1050,837,3,1,0.0029,0.33,0.0938
V,U450G287,1050,1930,5,3,0.0048,0.6,0.125
V,U450G287,1050,1939,1,0,0.001,0,0.0312
V,U450G843,700,827,1,1,0.0014,1,0.0312
```

2. **Conflating Trigger vs. Disposition Logic (Z-Score Flaw):**
   - Since the threshold Z is tied strictly to a product's overall recovery rate, there is a blindspot: when a hardware failure strikes a traditionally "low recovery" product, it gets held. But because the product is deemed "low recovery" globally, the AI will likely decides to release the lot *without* retesting it -> Loses the chance to fix the hardware and retest the lot (which would have recovered ~93% of those units).
3. **Outlier Skew distorts μ and σ:**
   - K-lot distributions are inherently right-skewed with extreme outliers heavily impacting the mean and standard deviation.
   - *Example:* **RPR8161** has a mean of 0.0373 but a median of just 0.0271 (Skewness 7.37). The mean is inflated 38% above the actual center. This inflation creates artificially loose baseline thresholds across the board.

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3. Proposed Fixes & Advanced Logic

Overcoming the Flaws

Problem	Root Cause	Fix
Hardware failures not caught correctly	SBL only looks at klot %; `CA_pct` is not used as a primary gate before the AI steps in.	Add a **Hardware Failure Gate** to trigger hardware inspection and socket swaps *before* running AI analysis.
"Low recovery" prevents retests	Z-score ties generic recovery rate to limit triggers, which ignores localized causes (like hardware).	Separate the hardware gate from disposition logic so hardware breaks always force a retest.
Outlier skew on μ and avg recovery	Right-skewed klot distributions heavily distort `mean` and `std`.	Replace mean/std with **median/IQR**.

Redefined Trigger Threshold Formula to remove outlier(s) influence

The Formula:

$$T_p = \text{median}(\text{klot}) + (Z_{adj}) \times \frac{\text{IQR}(\text{klot})}{1.35}$$

Why IQR / 1.35?
Using the IQR bounds *filters outliers* while maintaining a mathematical link to standard deviation.
In a perfect Gaussian distribution

- The **25th percentile** ($Q1$) is at roughly **-0.6745** $\times\sigma$ from the mean.
- The **75th percentile** ($Q3$) is at roughly **+0.6745** $\times\sigma$ from the mean.

The IQR is the distance between $Q3$ and $Q1$:

$$\text{IQR} = Q3 - Q1$$

$$\text{IQR} = (0.6745 \times \sigma) - (-0.6745 \times \sigma)$$

$$\text{IQR} = 1.349 \times \sigma$$

$$=> \sigma \approx \frac{\text{IQR}}{1.35}$$

Parameter Mapping:

Component	Current (Baseline)	Recommended (Robust)
Center	`mean(klot)`	`median(klot)`
Spread	`std(klot)`	`IQR(klot) / 1.35`
Recovery Rate for Z	`mean(recovery_rate)`	`median(recovery_rate)`

Cost-Efficiency Ratio (CER)

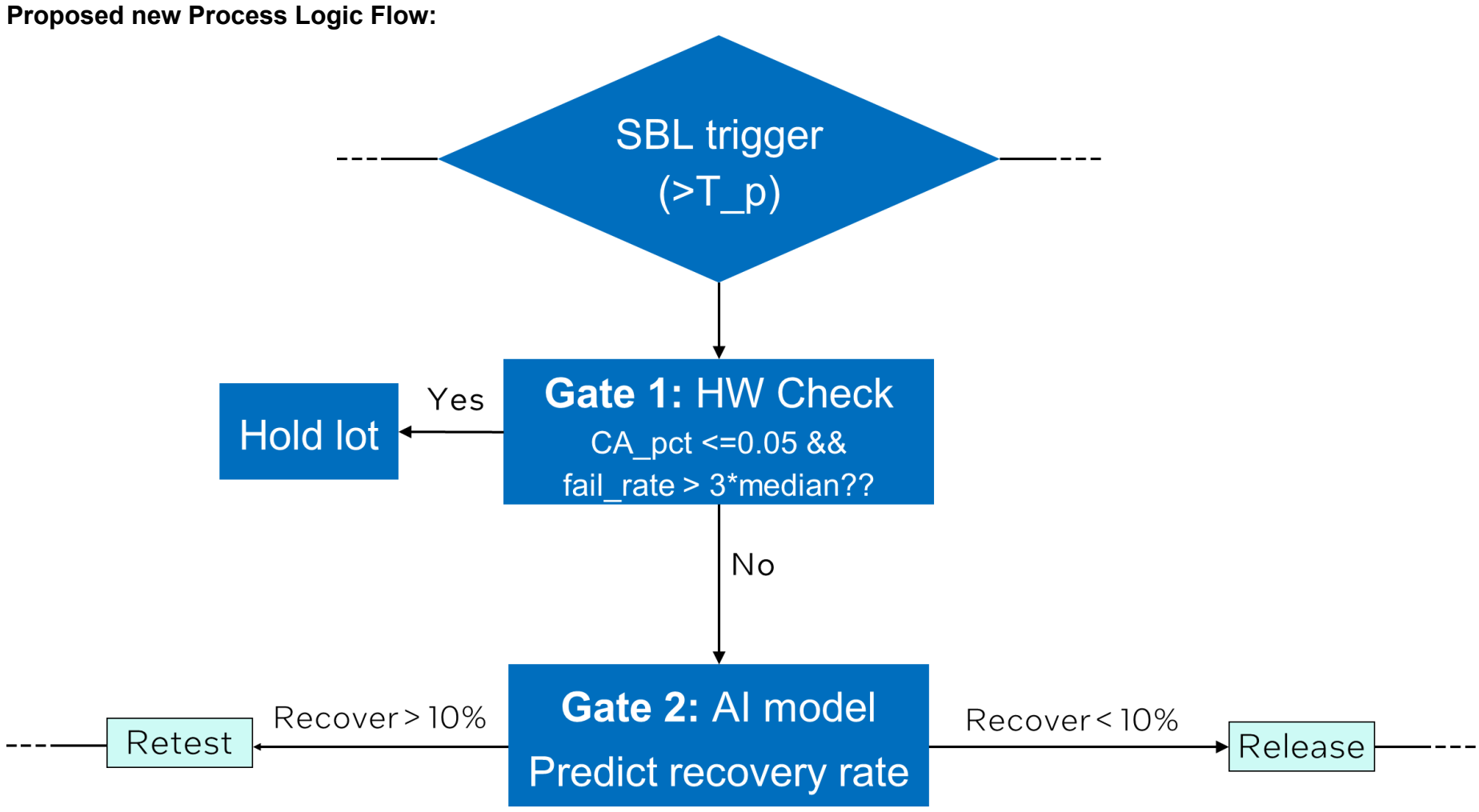
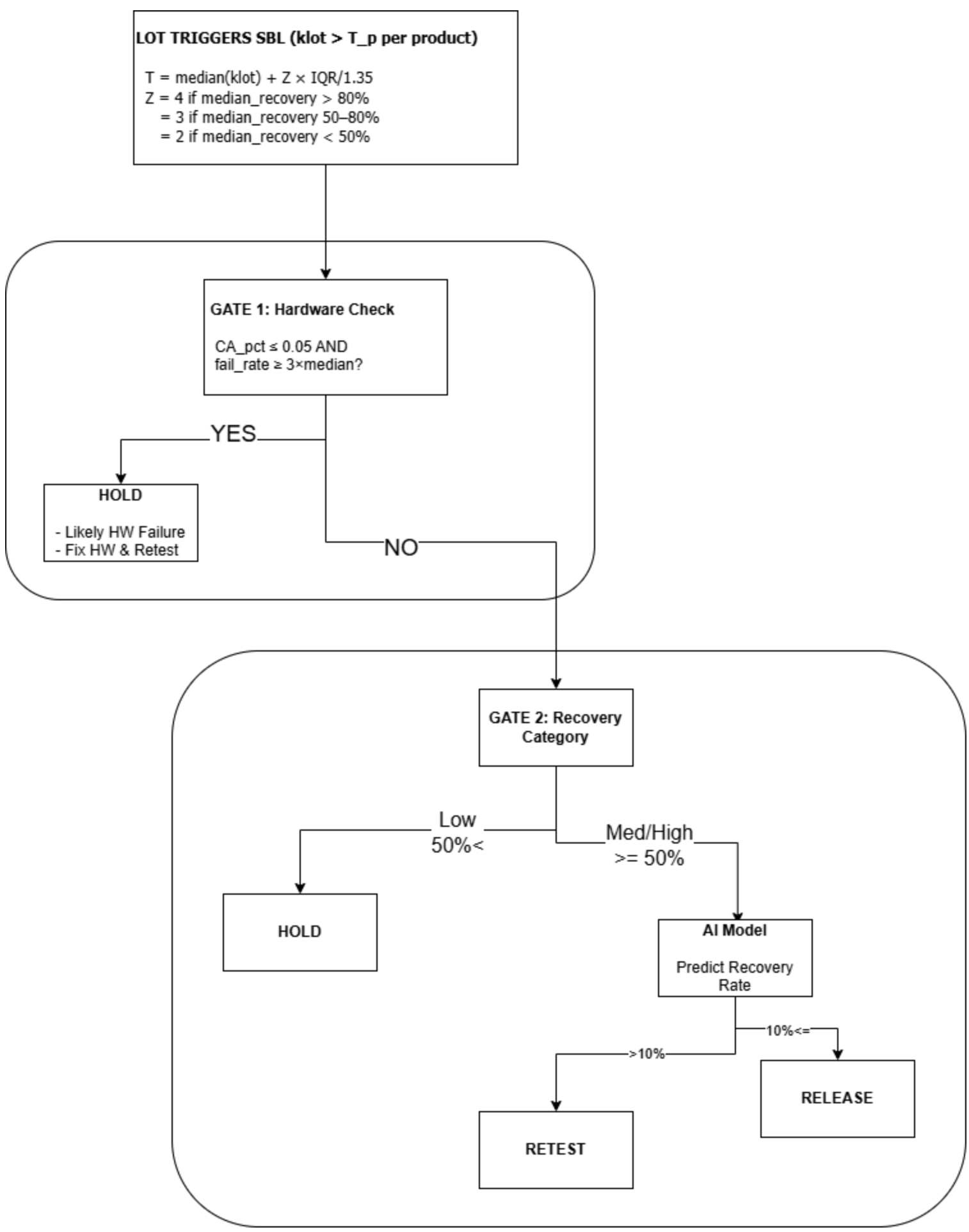
To account for the ECONOMIC impact of releasing vs retesting, an extra parameter is added to the Base Z-score calculation to get Z_{adj}.

$Z_{adj} = Z_{\text{median(recovery_rate)}} + \text{CER}_{adj}$

$$CER_{adj} = \frac{(potential_recovered_units * unit_yield_value - test_cost)}{test_cost}$$

```
# CER (Cost-Efficiency Ratio) calculation:
CER = (recovered_units * unit_yield_value - test_cost) / test_cost

if CER > CER_HIGH:
    # Retest is highly economical -> loosen threshold
    CER_adj = +0.5
elif CER < CER_LOW:
    # Retest is marginally or not worth it -> tighten threshold
    CER_adj = -0.5
else:
    CER_adj = 0
```

Integrated Workflow: SBL Hardware Gate

Proposed new Process Logic Flow: